

Tech-Edge Lab

Topic: -

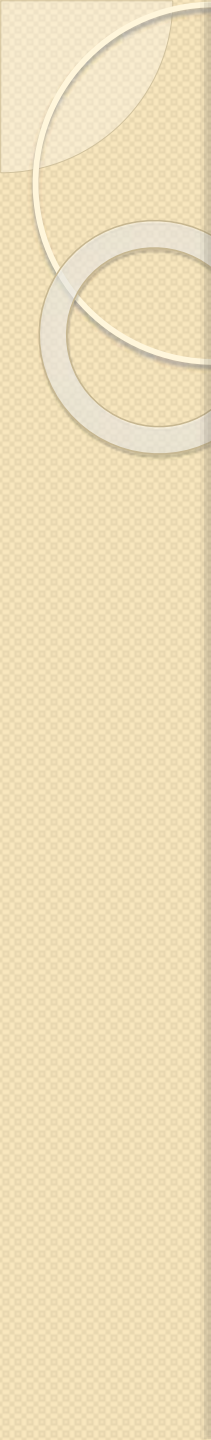
**Generative AI & Intelligent Application
Development Bootcamp**

Sub-topic:-

**Industry-Oriented Training Program on
LLMs, AI Tools, AI Agents & Vibe Coding**

Instructions to follow carefully:-

- Be curious and ready to learn beyond the classroom.
- Bring your laptop and ensure a stable internet connection for practice.
- Attend every session and participate actively.
- Stay focused during class and complete all hands-on activities.
- Practice regularly after each session.
- Ask questions whenever you have doubts.
- Work on mini-projects to strengthen your skills.
- Maintain discipline and a positive learning attitude.



Module 1:- AI industry and Careers

Lecture 1:-

Introduction to AI Era

(Course Introduction, Why learn AI, AI Vs Programming, AI Jobs, AI salary, AI roadmap for BCA, What companies expect)

Activity:

Raise Your Hand Survey

Question:

How many students have already used ? :-

- ChatGPT
- Gemini
- GitHub Copilot
- Claude
- Perplexity

Q- "Who has used AI for completing assignments?"

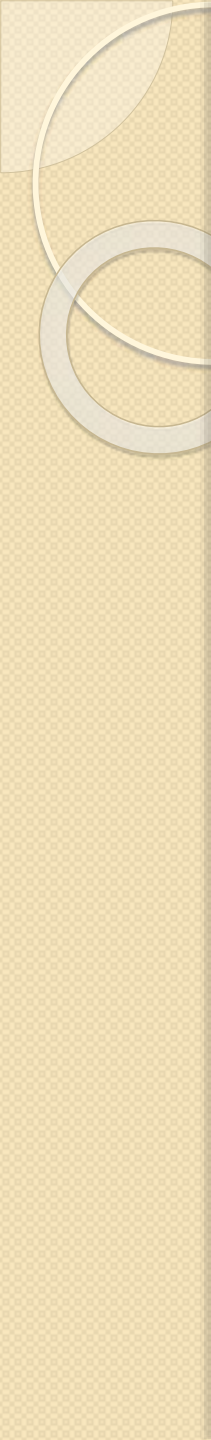
Q- "Who has used AI for coding?"

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1. Introduction

- **What is this Bootcamp?**
 - This bootcamp is designed to prepare students for the rapidly changing software industry where Artificial Intelligence is becoming a core part of software development.
 - The program focuses on practical learning rather than theory. Students will learn how to use modern AI tools to increase productivity, automate tasks, and build intelligent applications.



What You Will Learn

- Fundamentals of Artificial Intelligence
- Generative AI
- Large Language Models (LLMs)
- AI Chatbots
- AI Coding Assistants
- Prompt Engineering
- AI Agents
- Intelligent Web Applications
- API Integration
- Vibe Coding
- AI Project Development
- Resume Building using AI
- Industry Workflow

2. Why This Bootcamp?

- Today's software companies no longer expect developers to write every line of code manually.
 - Instead, they expect developers who can:
 - Use AI tools efficiently
 - Build applications faster
 - Automate repetitive tasks
 - Improve software quality
 - Increase productivity
 - Solve real-world problems using AI
- This bootcamp bridges the gap between college education and industry expectations.

3. Learning Outcomes

- After completing this bootcamp, students will be able to:-
 - ✓ Understand AI fundamentals
 - ✓ Use ChatGPT effectively
 - ✓ Work with multiple AI tools
 - ✓ Build AI-powered applications
 - ✓ Use LLM APIs
 - ✓ Create AI Agents
 - ✓ Perform Prompt Engineering
 - ✓ Develop projects faster using AI
 - ✓ Prepare for AI-enabled software jobs

ACTIVITY: IDENTIFY THE AI TOOLS

Look at the logos below and identify the AI tools.

1



2



3



4



5



Midjourney

6



perplexity



TIP: Think about the AI tools used for chatting, writing, coding, image generation, research, and more!

4. Welcome to the AI Era

- The Fourth Industrial Revolution
 - Earlier Revolutions
 - Steam Engine
 - Electricity
 - Computers & Internet
 - Artificial Intelligence
- Today, AI is transforming every industry just like computers transformed businesses in the 1990s.
 - Examples
 - Healthcare
 - Banking
 - Education
 - Agriculture
 - Manufacturing
 - Retail
 - Government
 - Entertainment
 - Cyber security

5. AI is Everywhere

- Examples of AI You Already Use
 - Google Search
 - YouTube Recommendations
 - Netflix Suggestions
 - Instagram Feed
 - Amazon Recommendations
 - Google Maps
 - ChatGPT
 - GitHub Copilot
 - Google Gemini
 - Claude
 - Perplexity
- AI has become part of our daily life—even when we don't realize it.

Activity: AI Hunt

Q-

Find 10 AI applications you used today before entering class.

Examples

?

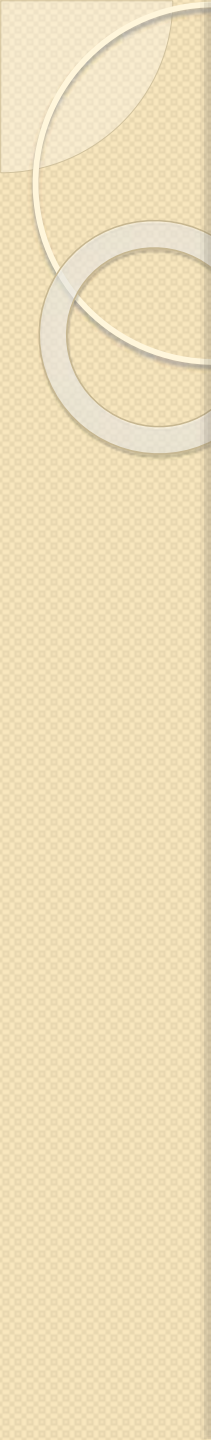
Give 2 minutes.

6. Why Learn AI?

According to industry reports,
AI is becoming one of the fastest-growing technologies.

Reasons to Learn AI

- High demand in software companies
- Better salary packages
- Faster career growth
- Increased productivity
- Ability to automate work
- Competitive advantage
- Better freelancing opportunities
- Startup opportunities



Activity Ask?

Why are you learning AI?

Options

- A. Placement
- B. Higher Salary
- C. Freelancing
- D. Startup
- E. Curiosity

Activity:- Think-Pair-Share

Question

Imagine you get placed in TCS.

Would you rather

A)

Write 1000 lines manually

OR

B)

Use AI to finish in 2 hours

7. Why AI Matters for BCA Students

- Traditional Skill

Write code manually



Modern Skill

Use AI + Programming



Future Skill

- Build AI-powered products
- Companies now expect graduates to know both programming and AI.

8. AI vs. Traditional Programming

Traditional Programming	AI Programming
Human writes logic	AI generates logic
Manual coding	AI-assisted coding
Time consuming	Faster development
Fixed rules	Learns patterns
Programmer solves everything	AI assists developer
Less automation	High automation

Important:

AI does NOT replaces programming.

AI enhances programming.

9. Can AI Replace Programmers?

Common Myth

"AI will replace software engineers."

Reality

AI replaces repetitive tasks—not problem-solving skills.
Companies still need developers who can:

- Think logically
- Design software
- Understand requirements
- Debug systems
- Communicate with clients
- Review AI-generated code

The future belongs to developers who know how to use AI effectively.

Activity :- Mini Challenge

Ask

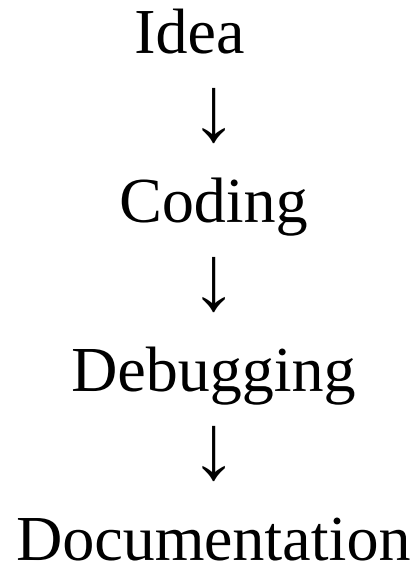
Can AI solve these without humans?

- Decide business requirements?
- Understand customer emotions?
- Handle office politics?
- Design software architecture?
- Fix production bugs?

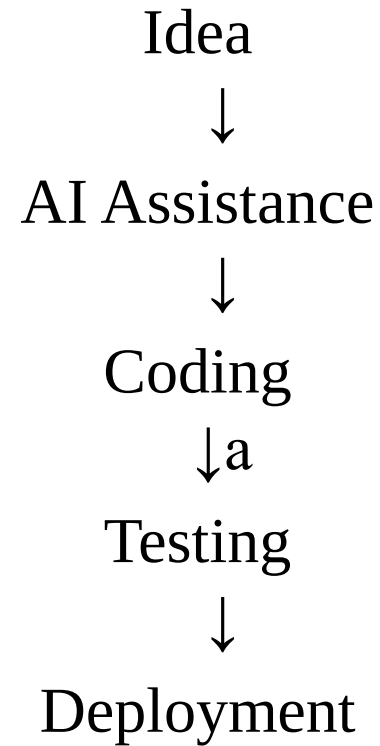
- Yes/No.

10. AI + Programmer = AI Enabled Programmers

- Without AI



- With AI



- Result

- Faster Development
 - Better Productivity
 - Higher Quality

Demonstration

A student

"Write code to reverse a string."

B student

Use ChatGPT.

Compare

- Time
- Accuracy
- Readability

11. Evolution of Software Development

- Past

Write every line manually.

- Present

AI suggests code.

- Future

Describe your idea.

↓

AI generates the application.

↓

Developer verifies and improves it.

- This is called **AI-assisted software development**.

11. What is Generative AI?

Generative AI is a type of Artificial Intelligence that creates new content instead of only analyzing existing data.

It can generate

- Text
- Images
- Videos
- Audio
- Code
- Documents
- Presentations
- Websites

Examples

- ChatGPT
- Gemini
- Claude
- Midjourney
- GitHub Copilot

Activity:- Live Demo

Open ChatGPT.

Ask students

Give me one random prompt.

Examples

- Write a leave application
- Create logo
- Python program
- Explain DBMS

Generate instantly.

Huge engagement.

12. AI Jobs in 2026

Popular Roles

- AI Engineer
- Prompt Engineer
- Machine Learning Engineer
- Data Scientist
- AI Application Developer
- Full Stack AI Developer
- LLM Engineer
- AI Automation Engineer
- AI Product Manager
- AI Research Engineer
- AI Consultant

13. AI Skills Required by Companies

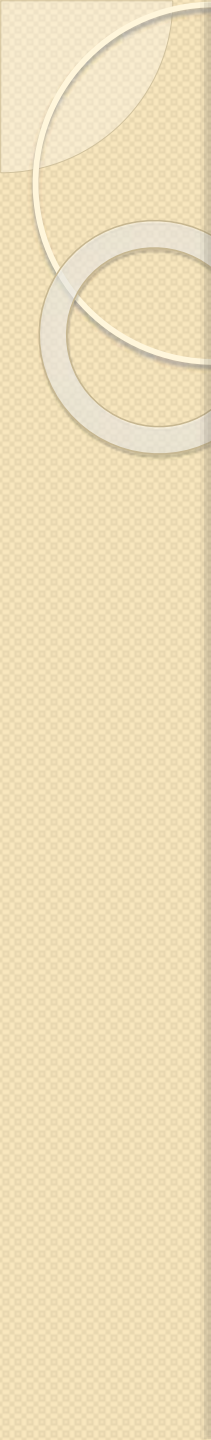
- Problem Solving
- Programming
 - Python
 - JavaScript
 - Java
- AI Skills
 - Prompt Engineering
 - LLMs
 - AI APIs
 - Automation
- Tools
 - Git
 - GitHub
 - VS Code
- Cloud
- Communication
- Projects , Portfolio

14. AI Salary (India)

- Approximate Annual Salary
 - Freshers
 - ₹3 –10 LPA
 - 1–3 Years
 - ₹10–20 LPA
 - Senior AI Engineer
 - ₹20–40+ LPA
 - Top Product Companies
 - ₹40 LPA to ₹1 Crore+ (Total Compensation)
- **Note:** Salaries vary based on skills, company, location, and experience.

15. AI Roadmap for BCA Students

- Step 1
Programming Fundamentals
↓
- Step 2
Python
↓
- Step 3
Data Structures
↓
- Step 4
Web Development (SDLC)
↓



- Step 5
Git & GitHub



- Step 6
Generative AI



- Step 7
LLMs



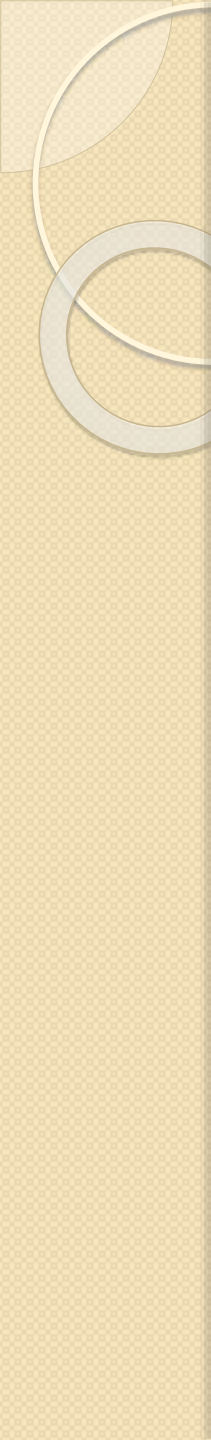
- Step 8
Prompt Engineering



- Step 9
AI APIs



- Step 10
AI Agents

- 
- Step 11
Projects



- Step 12
Internship & Placement

16. What Companies Expect

i) Technical Skills

- Programming
- Database
- APIs
- Git
- AI Tools
- Cloud Basics

ii) Professional Skills

- Communication
- Teamwork
- Critical Thinking
- Problem Solving
- Adaptability
- Continuous Learning

iii) AI Skills

- ChatGPT , Prompt Engineering
- AI Productivity, LLM Integration, AI Automation

17. Future Software Developer

- Traditional Developer

Writes Code



- Modern Developer

Writes Code + Uses AI



- Future Developer
 - Builds Intelligent Applications
 - Creates AI Agents
 - Automates Business Processes

18. Bootcamp Roadmap

Module 1

Intro to AI Era

Module 2

Prompt Engineering

Module 3

AI Tools

Module 4

Large Language Models

Module 5

OpenAI APIs

Module 6

Building AI Applications

Module 7

AI Agents

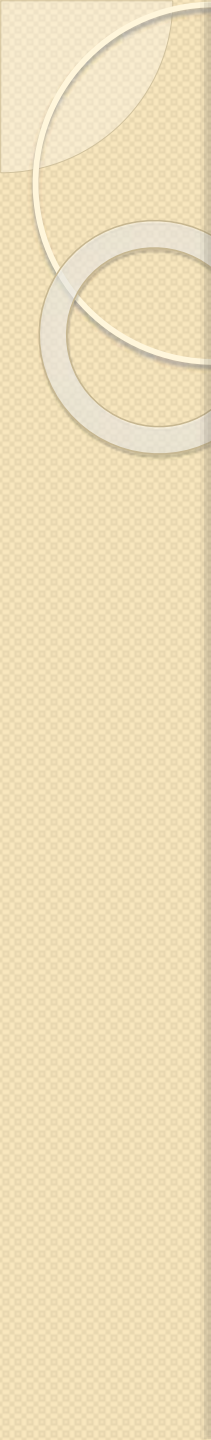
Module 8:- Vibe Coding

Module 9:- Project Development

Module 10:- Resume + Interview Preparation

19. Key Takeaways

- AI is transforming every industry.
- AI is a powerful assistant, not a replacement for skilled developers.
- Generative AI creates text, code, images, audio, and more.
- BCA students should combine programming skills with AI knowledge.
- Companies increasingly expect graduates to know AI tools, LLMs, and prompt engineering.
- Continuous learning and hands-on projects are essential for success in the AI era.



Thank You

Questions & Discussion